

posterior end of the alveolar process, is a synapomorphy of the ingroup (Heroini). Within Heroini, there is a transition to State 0 in Clade D. Within Clade D, there is a transition to State 1 in *Archocentrus* and in Clade G. Within Clade G, there is a transition to State 0 in Clade K. Within Clade K, there is a transition to State 1 in *Thorichthys*. (ci=.17, ri=.70)

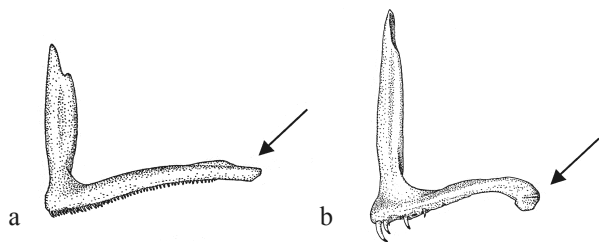


Fig. 5. Character 4, premaxilla, lateral view. **a**, condition 0, *Cichla ocellaris*, UMMZ 215935; **b**, condition 1, *Parachromis managuense*, UMMZ 223246.

5. Mental prominence on dentary (Fig. 6)

- 0 : weak,
1 : strong

Barel *et al.* (1977) defined mental prominence as a protuberance of the dentary symphysis. State 1, a distinguished mental prominence, is "when the ventral side of the dentary near its symphysis forms a ventral projection which clearly interrupts the main course of its ventral border" (Barel *et al.*, 1977). The transition to State 1, presence of a strong mental prominence on the dentary, is a synapomorphy of Clade A. Within Clade A, a transition to State 0 is present in "*Cichlasoma*" *salvini*, Clade G, and Clade R. Within Clade G, there is a transition to State 1 in "*Cichlasoma*" *urophthalmum*. Within Clade R, there is a transition to State 1 in *Nandopsis*. (ci=.17, ri=.74)

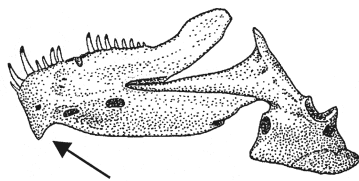


Fig. 6. Character 5, lower jaw, lateral view, condition 1, *Parachromis friedrichsthalii*.

6. Shape of lateral face of retroarticular (Fig. 7)

- 0 : rectangular,
1 : triangular

The transition to State 1, having a triangular-shaped retroarticular, is a synapomorphy of Clade C. There is a transition to State 0 in "*Cichlasoma*" *ornatum*, "*C.*" *istlanum*, and "*C.*" *urophthalmum*. (ci=.25, ri=.70)

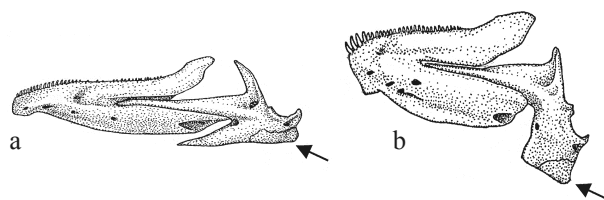


Fig. 7. Character 6, lower jaw, lateral view. **a**, condition 0, *Cichla ocellaris*, UMMZ 204679; **b**, condition 1, *Nandopsis tetracanthus*, UMMZ 146972.

7. Retroarticular and process of anguloarticular on same vertical plane (Fig. 8)

- 0 : no,
1 : yes

The transition to State 1, having the retroarticular and process of anguloarticular on the same vertical plane, is a synapomorphy of Clade R, and Clade F. Within Clade F, there is a transition to State 0 in *Thorichthys*, "*Cichlasoma*" *facetum*, and "*Cichlasoma*" *trimaculatum* + "*C.*" *urophthalmum*. (ci=.17, ri=.69)

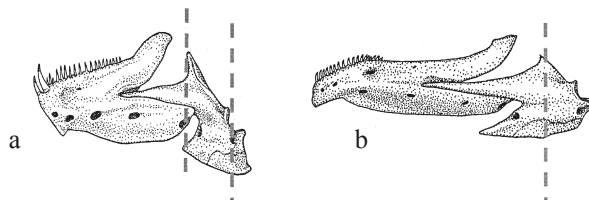


Fig. 8. Character 7, lower jaw, lateral view. **a**, condition 0, "*Cichlasoma*" *salvini*, UMMZ 188062-5; **b**, condition 1, *Crenicichla saxatilis*, UMMZ 215935.

8. Angle of process of anguloarticular (Fig. 9)

- 0 : procumbent,
1 : erect or recurved

There is a transition to State 1, having an erect or recurved anguloarticular process, in "*Cichlasoma*" *trimaculatum*, *Amphilophus macracanthus*, *Thorichthys meeki* + *T. pasionis*, "*Cichlasoma*" *grammodes*, Clade R, and Clade M. Within Clade R, there is a transition to State 0 in *Nandopsis haitiensis*. Within Clade M, there is a transition to State 0 in *Vieja fenestrata* + *V. maculicauda* + *V. synspila*. This character is similar to Character 8 of Cichocki (1976); however, he quantified the measure into five states based on angles between the posterior margin of the retroarticular process of the mandible and the mandibular axis. In my study, intraspecific variation precluded finding similarly discrete quantitative character-states. (ci=.13, ri=.59)